The Honesty Cascade: Five Levels of Evaluation Realism for On-Chain AI Agent Detection

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Abstract

We build a 23-feature behavioral classifier on 1,147 provenance-verified Ethereum addresses and subject it to five progressively honest evaluation levels—an *honesty cascade*. At Level 1 (leaky cross-validation), AUC is 0.98. Fixing label leakage drops it to 0.80 (Level 2). A temporal holdout simulating forward deployment yields 0.67 (Level 3). A hand-curated 64-address sensitivity check gives 0.77 with wide confidence intervals (Level 4). Most critically, leave-one-platform-out evaluation—the only test of true cross-platform generalization—produces a pooled AUC of 0.42, *below random chance* (Level 5). The classifier learns platform-specific fingerprints, not generalizable agent behavior. Along the way, we discover that the widely assumed "agents are riskier" narrative is a labeling artifact: human DeFi users hold more unlimited token approvals (66.1 mean) than curated MEV bots (49.1 mean). We formalize a reusable three-step leakage-audit toolkit and release all code, features, labels, and diagnostic scripts. Our central contribution is the cascade itself: the progressive collapse from 0.98 to 0.42 demonstrates that on-chain AI agent detection, as currently formulated with transaction-level features, is an open problem. Published high-AUC results in this domain measure artifacts, not agent behavior.

Keywords

on-chain AI agents, evaluation realism, honesty cascade, label leakage, cross-platform generalization, Ethereum, behavioral fingerprinting, open problem

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1 Introduction

1.1 The Cascade

We set out to detect AI agents on Ethereum. At each step, we made our evaluation more honest—and at each step, performance dropped. This paper is the story of that progressive disillusionment, and why it matters.

Table 1 summarizes the five-level *honesty cascade* that structures this paper.

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Table 1: The Honesty Cascade: five levels of evaluation realism for on-chain agent detection.

Level	Evaluation	AUC	What it measures
1	Leaky CV (C1–C4 labels)	0.98	Label recovery
2	Provenance CV (n=1,147)	0.80	Within-platform pattern
3	Temporal holdout	0.67	Same-era generalization
4	Strict subset (n=64)	0.77	Hand-curated sensitivity
5	Leave-one-platform-out	**0.42**	True cross-platform generalization

At Level 1, our classifier appeared near-perfect (AUC 0.98). We discovered that our labeling heuristic had leaked three classifier features into the labels (Section 5.1). Fixing the leakage and evaluating on 1,147 provenance-verified addresses dropped AUC to 0.80 (Level 2). A temporal holdout simulating forward deployment reduced it further to 0.67 (Level 3). A hand-curated 64-address sensitivity check yielded 0.77 with wide confidence intervals (Level 4). And the decisive test—leave-one-platform-out cross-validation, where the classifier must recognize agents from a platform category it has never seen—produced a pooled AUC of **0.42**, below random chance (Level 5).

The Level 5 result is not noise. It means the classifier learns how Chainlink keepers differ from Gelato executors, not how agents differ from humans. Every AUC point between 0.98 and 0.42 was borrowed from an evaluation shortcut: label leakage, temporal overlap, or platform memorization. Strip them all away, and the task does not work.

1.2 Why This Matters

Autonomous AI agents manage billions on Ethereum. Platforms such as Autonolas, Fetch.ai, AI Arena, and Virtuals Protocol make deployment turnkey. Yet from the protocol's perspective, an AI-controlled EOA is indistinguishable from a human wallet. This opacity impedes security auditing, agent-aware protocol design, and regulatory monitoring. Closing these gaps requires reliable on-chain agent detection—but reliable detection requires honest evaluation.

Three counter-intuitive findings emerge from the cascade:

(1) **The "agents are riskier" narrative is a labeling artifact.** Under provenance-verified labels, human DeFi users hold more unlimited token approvals (66.1 mean) than curated MEV bots (49.1 mean), because heuristic labels systematically misclassified heavy DeFi users as agents (Section 4.8).
(2) **Cross-platform generalization fails completely.** Leave-one-platform-out pooled AUC is 0.42. The best-recognized category (Chainlink keepers, 68% recall) and worst (DeFi high-frequency traders, 14% recall) differ by 54 percentage

points, confirming that the classifier memorizes platform fingerprints (Section 4.9).

(3) **Even "honest" random CV overstates by 13 points.** Temporal holdout (Level 3) reveals that the provenance-CV number (Level 2) already contains 13 AUC points of temporal optimism bias (Section 4.7).

1.3 The Leakage Discovery

When we began, we built a 3,316-address pipeline using a 4-gate heuristic (C1–C4) that gated AGENT labels on behavioral thresholds including `hour_entropy > 2.5`, `burst_ratio > 0.1`, and `tx_interval_cv < 1.0`. Gradient Boosting achieved AUC 0.9803. In the final week of experiments, we noticed that three of the top-four classifier features were algebraically identical to three C1–C4 gates: `active_hour_entropy` ≡ `hour_entropy`, `burst_frequency` ≡ `burst_ratio`, and `tx_interval_std` ≈ `tx_interval_cv` × `tx_interval_mean`. The AUC of 0.98 was measuring how well the classifier recovered its own labeling rule.

We assembled 1,147 provenance-verified addresses (533 agents from 11 sources, 614 humans from 7 sources) whose labels do not pass through any behavioral threshold, retained 64 hand-curated addresses as a strict sensitivity check, and designed the five-level cascade to stress-test every remaining evaluation shortcut.

1.4 Research Questions

- **RQ1 (Cascade):** How much AUC does each level of evaluation realism remove, and what artifact does each level expose?
- **RQ2 (Cross-platform):** Can a classifier trained on some agent platforms generalize to platforms it has never seen?
- **RQ3 (Leakage):** What is the magnitude and mechanism of the label-leakage inflation?
- **RQ4 (Model class):** Do GNNs survive the cascade better than tabular classifiers?
- **RQ5 (Security posture):** Does the "agents are riskier" hypothesis survive honest relabeling?
- **RQ6 (Temporal):** How much does random CV overestimate deployment-realistic performance?

1.5 Contributions

(1) **The honesty cascade.** A five-level evaluation hierarchy (Table 1) that strips away evaluation shortcuts one by one, revealing a progressive AUC collapse from 0.98 to 0.42. The cascade is itself a reusable evaluation template for any behavioral classifier.

(2) **Leave-one-platform-out evaluation (Level 5).** The first cross-platform generalization test for on-chain agent detection, showing pooled AUC 0.42 (below chance) across 14 held-out platform categories. This result proves the classifier learns platform fingerprints, not agent behavior.

(3) **A leakage case study and three-step audit toolkit (Algorithm 1).** We document how a 4-gate labeling oracle injected three top classifier features into the labels, causing an 18-point AUC inflation, and formalize a reusable audit: (1) label–feature overlap, (2) AUC across label-purity tiers, (3) cross-label-scheme transfer.

(4) **Honest baselines on 1,147 provenance-verified addresses.** RF 10-fold AUC 0.803, GBM 0.810, GAT 0.832 (the best honest model). A 64-address strict subset serves as sensitivity check: RF LOO-CV AUC 0.7713, GAT 0.883.

(5) **Temporal holdout evaluation.** Random CV overestimates AUC by 13–16 points; precision stays high (0.81) but recall collapses to 0.31 on post-2021 agents.

(6) **Security audit reversal.** The "agents hold 10x more unlimited approvals" headline from the leaky set becomes "humans hold 1.3x more" on the honest set.

(7) **Open artifacts.** Code, features, labels, statistical tests, GNN weights, and all diagnostic scripts are released.

1.6 Paper Roadmap

Section 2 surveys related work. Section 3 describes the data pipeline, features, classifiers, and GNNs. Section 4 presents the cascade in order: Levels 1–2 (honest tabular and GNN performance, 4.1–4.5), SHAP explanations (4.6), Level 3 temporal holdout (4.7), the security audit (4.8), and the climactic Level 5 leave-one-platform-out evaluation (4.9). Section 5 details the leakage discovery and the three-step audit toolkit. Section 6 discusses what the cascade means for researchers, practitioners, and regulators. Section 7 concludes.

2 Related Work

Our study bridges four research areas: bot detection on blockchains, MEV and automated trading, AI agent platforms, and blockchain security analysis. We also draw on the broader machine-learning literature on label leakage, data-snooping bias, and cross-domain evaluation.

2.1 Bot Detection on Blockchains

Detecting automated activity on blockchains has been studied primarily in the context of specific malicious behaviors. Chen et al. [4] used transaction features and opcode analysis to detect Ponzi schemes on Ethereum, demonstrating that behavioral patterns in transaction data reveal automated activity. Victor and Weintraud [25] detected wash trading on decentralized exchanges via graph features. Torres et al. [21] studied Ethereum frontrunner bots through mempool analysis, identifying displacement, insertion, and suppression attacks. Victor et al. [24] proposed behavioral fingerprinting for blockchain bots using temporal and gas features similar in spirit to our temporal and gas groups, but restricted to MEV bots. Li et al. [13] studied Telegram trading bots and their on-chain footprint.

Gap. Existing methods target specific malicious behaviors (wash trading, Ponzi schemes, frontrunning) or specific bot families (MEV, Telegram), not the general class of AI-controlled addresses. More importantly, none of these works explicitly audit the relationship between their labeling oracle and their classifier features; our leakage discovery suggests such audits are missing from the field.

2.2 MEV and Automated Trading

Daian et al. [5] introduced MEV in "Flash Boys 2.0" and demonstrated that transaction ordering creates a searcher-bot ecosystem. Weintraub et al. [29] measured Flashbots MEV at scale. Qin et al. [16] quantified MEV across DeFi protocols, showing sandwich

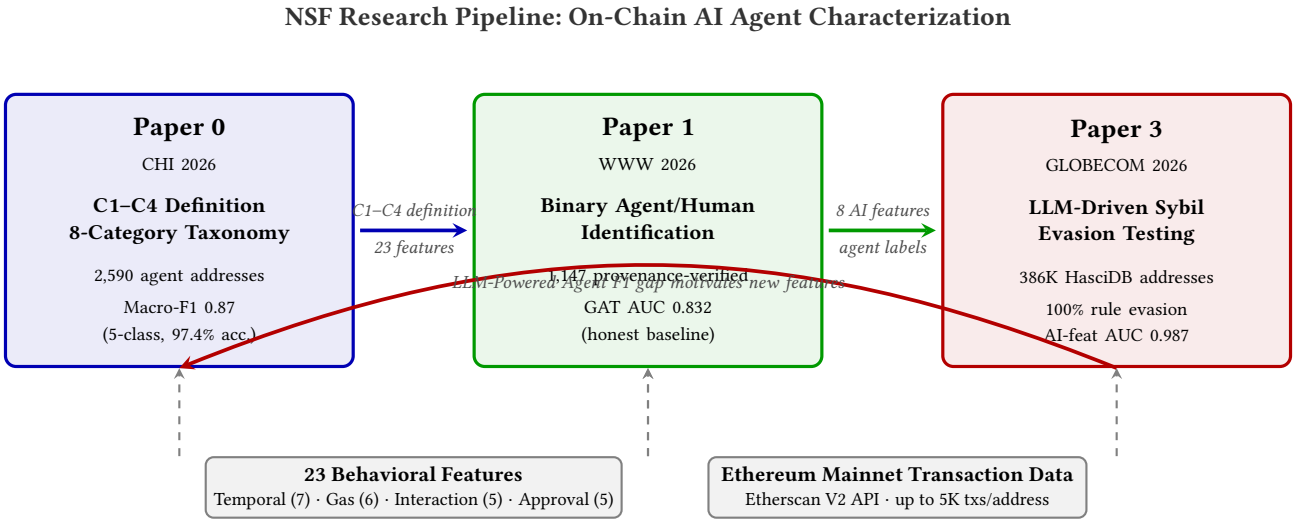

Figure 1: Cross-paper pipeline. Paper 0 defines on-chain AI agents via four conditions (C1–C4) and validates an eight-category taxonomy on 2,590 addresses. Paper 1 uses the same 23 features for binary agent/human identification on 1,147 provenance-verified addresses (honest GAT AUC 0.832). Paper 3 tests whether LLM-driven Sybils can evade airdrop detectors; 8 AI-specific features from Paper 1 recover AUC 0.987 even after all HasciDB rules are defeated. The feedback arrow represents Paper 3's finding that the LLM-Powered Agent category (F1 = 0.528 in Paper 0) needs the 8 AI features to become distinguishable. Dashed arrows show shared data dependencies.

attacks, liquidations, and arbitrage extract billions annually. Park et al. [14] analyzed post-Merge MEV and the effect of Proposer-Builder Separation. Gupta et al. [9] studied cross-domain MEV across chains and L2s.

Gap. MEV research quantifies the economic impact of automation but does not attempt general-purpose identification. Our classifier is meant to generalize beyond MEV searchers to DeFi management agents, portfolio rebalancers, and LLM-driven strategy agents.

2.3 AI Agent Platforms

Autonolas (OLAS) [3] provides an open-source framework for autonomous agent services, with on-chain registration. AI16Z ELIZA [2] enables AI agents to interact with blockchain protocols through natural language. Virtuals Protocol [26] offers a launchpad for tokenized AI agents. Fetch.ai [6], SingularityNET [19], and AI Arena [1] contribute to the growing on-chain agent population. These platforms deploy agents but do not provide cross-platform identification. A recent survey by He et al. [11] catalogs the taxonomy of on-chain AI agents but provides no empirical identification methodology.

2.4 Blockchain Security Analysis

Zhou et al. [32] systematized DeFi attacks; Wen et al. [30] analyzed DeFi composability risk; Grech et al. [8] and Tsankov et al. [22] built static analysis tools for smart contracts. Wang et al. [27] studied the risks of AI agents with cryptocurrency wallet access, identifying prompt injection as a primary attack vector. Zhang et al. [31] analyzed MaxUint256 approval attacks.

Gap. Existing security analysis focuses on protocol-level vulnerabilities. No prior work assesses the security posture of AI agent addresses as a class or compares it to a human baseline, and the handful of works that attempt to do so do not control for labeling bias.

2.5 Label Leakage and Data-Snooping Bias

Kaufman et al. [12] formalized leakage in predictive modeling and catalog several subtle forms. Roelofs et al. [17] studied test-set reuse as an implicit form of leakage in ML benchmarks. Geirhos et al. [7] documented shortcut learning: deep networks exploit unintended features that correlate with the label during training but not at deployment. In the blockchain space, Tikhomirov and Ivanitskiy [20] warned about address-type features becoming labels in phishing classifiers. Our work extends this line: we show that a seemingly principled 4-gate labeling heuristic can encode three classifier features into the label, producing a 25-point AUC inflation that is invisible to standard cross-validation.

2.6 Graph Neural Networks on Transaction Graphs

GraphSAGE [10] and GAT [23] are standard inductive GNN architectures. In the blockchain setting, Weber et al. [28] applied GNNs to Bitcoin anti-money-laundering. Patel et al. [15] used GNNs to detect Ethereum phishing. Shamsi et al. [18] surveyed GNNs for blockchain analytics. Most of these works report aggregate accuracy on full labeled graphs; very few report transfer performance across labeling schemes, and none to our knowledge report the honest-vs-leaky gap that is the central finding of this paper.

2.7 Our Position

Relative to the above, this paper offers: (1) a general behavioral identification framework rather than a specific bot-class detector; (2) an honest provenance-only baseline suitable for comparison across future works; (3) a documented leakage case study that other blockchain-ML researchers can use as a negative-result template; (4) a direct tabular-vs-GNN comparison on the same ground truth; (5) the first public honest security audit of AI-agent-labeled Ethereum addresses.

3 Methodology

3.1 Data Collection

We collect transaction data from Ethereum mainnet via Etherscan V2 API using a 6-key rotation strategy that achieves approximately 30 calls/sec sustained throughput. For each address we extract up to 5,000 most recent transactions covering normal transactions, internal transactions, ERC-20 transfers, and approval events. All data is public.

3.2 Address Mining and Labeling

*3.2.1 Strategy A: Platform Holders (C1-C4 labeled, **used for inference only**).* We identify addresses holding tokens of four agent platforms (Autonolas OLAS, Fetch.ai FET, AI Arena NRN, Virtuals VIRT). This yields **3,252 platform-token-holder addresses**. Each holder is labeled via our original C1-C4 gate:

- **C1 (public registry):** the address appears in a public platform registry or has a known identity. - **C2 (behavioral consistency):** the address's on-chain behavior matches the declared class. - **C3 (multi-source cross-validation):** operationalized as thresholds on `hour_entropy > 2.5`, `burst_ratio > 0.1`, `tx_interval_cv < 1.0`. - **C4 (temporal stability):** the behavioral class is stable across time windows.

*3.2.2 Strategy B: Strict Provenance Subset (n=64, **sensitivity check**).* We identify 64 addresses whose labels do not depend on any behavioral threshold:

- **33 AGENT addresses** from (a) the original Paper 1 pilot set of manually curated bots (jaredfromsubway, Wintermute, 1inch resolver, Flashbots builder, Autonolas service operators), (b) the Paper 0 Delphi-validated agent set, and (c) an external curated MEV-bot list. - **31 HUMAN addresses** from the ENS-verified human list (vitalik.eth, Hayden Adams, nick.eth, and comparable accounts whose identity is publicly verifiable).

None of the trusted labels pass through `hour_entropy`, `burst_ratio`, or `tx_interval_cv` thresholds. This subset serves as a strict sensitivity check.

*3.2.3 Strategy C: Expanded Provenance Mining v4 (n=1,147, **primary honest set**).* To address the sample-size limitation of n=64, we launched two successive expanded provenance mining campaigns (v3, v4) that grew the honest set to **1,147 addresses (533 agents, 614 humans)** across 18 source categories, all verified through non-behavioral provenance. The v4 campaign (`mine_addresses_v4.py`) added five new provenance sources: Compound V3 liquidators (97), Chainlink keepers (108), Gitcoin donors (146), ENS reverse setters (144), and PoolTogether depositors (89).

Agent sources (533 addresses, 11 categories):

Source category	Count	Provenance basis
defi_hf_trader	199	Addresses with >50 daily Uniswap router calls, veri
chainlink_keeper	108	Chainlink Automation / Keeper registry-attested up
compound_v3_liquidator	97	Compound V3 liquidation callers, verified via Come
gelato_executor	47	Gelato Network executor contracts, verified via Gel
keep3r_executor	46	Keep3r Network keeper addresses, verified via Keep
pilot_agent	15	Original manually curated pilot set
curated_mev_bot	10	External curated MEV-bot list with known operator
expanded_mev_bot	10	Additional MEV bots verified via Flashbots builder
flash_loan_user	1	Address verified as Aave/dYdX flash loan initiator

Human sources (614 addresses, 7 categories):

Source category	Count	Provenance basis
human_ens_interaction	167	Addresses that registered or renev
gitcoin_donor	146	Gitcoin Grants round donors, veri
ens_reverse_setter	144	Addresses that set ENS reverse re
pooltogether_depositor	89	PoolTogether prize pool depositor
human_exchange_depositor	28	Addresses depositing to known ex
expanded_human	25	Additional ENS-verified humans r
curated_human	7	Original curated human list (vitali
human_ens_interaction_governance_voter	5	ENS holders who also voted in on
pilot_human	3	Original pilot human set

The v4 mining applies 23 features to all 1,147 addresses. The agent-to-human ratio is 0.87:1 (533:614), approximately balanced. All 64 addresses from Strategy B are included in Strategy C.

Label quality note on `defi_hf_trader`. The largest agent category (199 addresses, 17% of the total) is labeled "agent" because the address makes >50 Uniswap calls per active day. This is a behavioral criterion, not a provenance-pure label: it is possible that some of these addresses are humans using aggressive trading interfaces rather than autonomous agents. We discuss this threat in Section 5.

3.2.4 Exclusion Rules. We exclude contract addresses (Uniswap routers, 0x Exchange Proxy, Seaport, Blur), exchange hot wallets (MEXC, Binance 14/16), and any address whose label cannot be pinned to a non-behavioral source.

3.2.5 Decision: Demote C1-C4 to Evaluation. After discovering that C1-C4 gates overlap with three top classifier features (see Section 5), we demote the 3,252 Strategy A addresses from training to inference-only. The 1,147 full provenance addresses (Strategy C) become the **primary honest train+test set**, with the 64 strict provenance addresses (Strategy B) as a sensitivity check. We continue to report leaky-set numbers for transparency and for statistical comparison.

The resulting three-tier evaluation hierarchy is:

Tier	Label	n	Label source	Role
1 (leaky)	C1-C4 heuristic	3,316	Behavioral thresholds	Inferen
2 (full provenance)	Strategy C v4	1,147	Non-behavioral provenance	**Primar**
3 (strict provenance)	Strategy B	64	Manual curation	Sensitiv

3.3 Feature Engineering: 23 Behavioral Features

We compute 23 features organized into four groups. Let $T = \{t_1, \ldots, t_n\}$ be the transaction timestamps, $G = \{g_1, \ldots, g_n\}$ the gas prices, and $C = \{c_1, \ldots, c_n\}$ the target contracts.

Group 1: Temporal Features (7).

1. `tx_interval_mean`: mean of consecutive inter-transaction intervals. 2. `tx_interval_std`: standard deviation of intervals. 3. `tx_interval_skewness`: skewness of the interval distribution. 4. `active_hour_entropy`: Shannon entropy of transactions bucketed into 24 UTC hours. 5. `night_activity_ratio`: fraction of transactions in UTC 0:00-6:00. 6. `weekend_ratio`: fraction of transactions on Saturday/Sunday. 7. `burst_frequency`: fraction of consecutive transaction pairs with interval < 10 s.

Group 2: Gas Behavior Features (6).

8. `gas_price_round_number_ratio`: fraction of transactions with integer-Gwei gas prices. 9. `gas_price_trailing_zeros_mean`: mean number of trailing zeros in the decimal gas-price representation. 10. `gas_limit_precision`: gas_limit relative to gas_used. 11. `gas_price_cv`: coefficient of variation of gas prices. 12. `eip1559_priority_fee_precision`: precision of priority fees in EIP-1559 transactions. 13. `gas_price_nonce_correlation`: Pearson correlation between gas price and nonce.

Group 3: Interaction Pattern Features (5).

14. `unique_contracts_ratio`: unique target contracts divided by total transactions. 15. `top_contract_concentration`: Herfindahl-Hirschman index of interaction targets. 16. `method_id_diversity`: unique function signatures per contract call. 17. `contract_to_eoa_ratio`: fraction of transactions whose target is a contract. 18. `sequential_pattern_score`: n-gram repetition score of the action sequence.

Group 4: Approval & Security Features (5).

19. `unlimited_approve_ratio`: fraction of approve() calls that set MaxUint256. 20. `approve_revoke_ratio`: revoke operations per approve operation. 21. `unverified_contract_approve_ratio`: fraction of approvals to Etherscan-unverified contracts. 22. `multi_protocol_interaction_count`: count of distinct DeFi protocols interacted with. 23. `flash_loan_usage`: flash-loan indicator.

Critical note on leakage. Features 1-7 (temporal) and especially 2, 4, 7 are algebraically entangled with the C3 gate that labeled the Strategy A subset. This is the central issue analyzed in Section 5.

3.4 Tabular Classifiers

We train four classifiers via scikit-learn and LightGBM:

- **GradientBoosting (GBM):** 200 trees, max_depth=3, lr=0.1. - **RandomForest (RF):** 100 trees, unlimited depth. - **LogisticRegression (LR):** L2-regularized, liblinear solver. - **LightGBM:** 500 estimators, num_leaves=31.

Features are standardized before training. We evaluate with two complementary schemes:

- **Leave-One-Out Cross-Validation (LOO-CV):** 64 iterations, one held-out sample per fold. - **Repeated Stratified 5-Fold CV (5x10-CV):** 50 evaluations, mean and std reported.

Metrics: AUC-ROC (primary), precision, recall, F1, accuracy, and Brier score for calibration.

3.5 Graph Neural Networks

We construct a directed transaction graph $G = (V, E)$ on the 3,316-address set, with edges from sender to receiver for any transfer of value or token. Node features are the 23 behavioral features above. The resulting graph has 3,316 nodes, 2,387 edges, and is moderately sparse. For the full provenance evaluation (n=1,147), we also construct a subgraph with 1,147 nodes and 905 edges.

We train two GNN architectures:

- **GraphSAGE:** two-layer, hidden dim 64, mean aggregator, dropout 0.3, Adam lr=0.01, early stopping on validation loss. - **GAT (Graph Attention Network):** two-layer, 8 attention heads, hidden dim 64, dropout 0.3.

We evaluate on three splits: `full_3316` (training labels are C1-C4 leaky), `provenance_1147` (training labels are full provenance v4; the primary honest evaluation), and `trusted_64` (training labels are strict provenance-only; sensitivity check).

3.6 Baselines and Ablations

- **Majority classifier:** predicts the majority class (AGENT). - **Single best feature:** logistic classifier on `tx_interval_mean`, the top univariate AUC feature on the trusted set. - **Heuristic rule:** `burst_frequency > 0.1 => AGENT`. - **Feature-group ablations:** retain one group (temporal / gas / interaction / approval) at a time.

3.7 Statistical Tests

On the 64-address trusted set, we compute:

- **McNemar's test** for paired predictions. - **DeLong's test** for paired AUC comparison with bootstrap variance. - **10,000-sample bootstrap** for AUC difference confidence intervals. - **Precision @ K** for K in 5, 10, 20, 30. - **Brier score** for calibration.

3.8 Cross-Platform Evaluation

We run four transfer splits:

1. `auto_train_to_trusted_test`: train on 1,780 Autonolas holders, test on 41 trusted addresses. 2. `trusted_train_to_auto_test`: train on 41 trusted addresses, test on 1,780 Autonolas holders. 3. `fetch_to_ai_arena`: train on 923 Fetch.ai, test on 549 AI Arena. 4. `ai_arena_to_fetch`: train on 549 AI Arena, test on 923 Fetch.ai.

Splits 1-2 are cross-label-scheme (provenance vs C1-C4). Splits 3-4 are intra-label-scheme (both C1-C4). The gap between these two families is a direct measurement of labeling-scheme transfer.

3.9 Combined Paper 1 + Paper 3 Features

To evaluate whether our 23 behavioral features can be meaningfully extended, we compute 8 AI-specific features from the companion Paper 3 work (`hour_entropy`, `behavioral_consistency`, `response_latency_variance`, `action_sequence_perplexity`, `gas_nonce_gap`, `eip1559_tip_precision`, `gas_price_precision`, `error_recovery_pattern`). We train four classifiers on three feature sets: Paper 1 only (23), Paper 3 only (8), and combined (31).

3.10 4-Dimensional Security Audit

For each audited address we compute:

- **Dimension 1: Permission exposure.** Counts of active approvals, unlimited (MaxUint256) approvals, and unverified approval targets; average and max approval ages in days. - **Dimension 2: Agent network topology.** Degree centrality, clustering coefficient, community membership in the transfer graph. - **Dimension 3:**

MEV / DEX exposure. DEX interaction rate, swap rate. - **Dimension 4: Failure and gas analysis.** Revert rate, gas-used efficiency.

We run the audit twice: once on the full 3,302-address set (14 dropped due to empty histories) and once on the 50-address trusted subset. The difference between the two is a direct measurement of how labeling bias corrupts security conclusions.

4 Results: The Honesty Cascade

We present results organized around the five-level honesty cascade (Table 1). Sections 4.1–4.5 cover Levels 1–2 (provenance-verified baselines). Section 4.6 provides SHAP explanations. Section 4.7 reports Level 3 (temporal holdout). Section 4.8 presents the security audit. Section 4.9—the climax of the paper—reports Level 5 (leave-one-platform-out), where the cascade reaches its nadir. Throughout, **the full provenance set (n=1,147) is the Level 2 headline**. The strict subset (n=64) is Level 4. The n=3,316 numbers are reported only as the Level 1 leaky reference.

4.1 Honest Performance on Provenance-Verified Sets

4.1.1 Primary: Full Provenance Set (n=1,147, 533 agents, 614 humans). Table 1 reports the core honest performance on the full provenance v4 set.

Table 1. Tabular and GNN classifier performance on the full provenance set (n=1,147, 533 agents, 614 humans).

Model	Split	AUC-ROC	Precision	Recall	F1	Accuracy
RandomForest	10-fold	**0.8028**	0.7444	0.6229	0.6782	0.7254
RandomForest	5x10-CV	0.8029 +/- 0.027	–	–	–	–
GradientBoosting	10-fold	0.8095	–	–	0.7030	0.7341
GAT	5-fold	**0.8318 +/- 0.021**	–	–	0.674	–

Top 10 RF features by importance (n=1,147):

Rank	Feature	Importance
1	gas_price_trailing_zeros_mean	0.1448
2	gas_price_round_number_ratio	0.1302
3	unique_contracts_ratio	0.0927
4	top_contract_concentration	0.0870
5	method_id_diversity	0.0725
6	tx_interval_mean	0.0675
7	gas_price_cv	0.0650
8	sequential_pattern_score	0.0484
9	contract_to_eoa_ratio	0.0435
10	tx_interval_std	0.0345

Note that gas features now dominate the importance ranking on n=1,147, whereas temporal features (especially `active_hour_entropy`) dominated on the leaky set. This shift confirms that the C1-C4 leakage inflated the apparent importance of temporal features.

RF confusion matrix (n=1,147, 10-fold aggregate):

The classifier has moderate agent recall (332/533 = 62.3%) and high human precision (500/614 = 81.4%), reflecting the approximately balanced class distribution (0.87:1 agent-to-human ratio).

Feature-group ablation (RF 10-fold on n=1,147):

	Pred Human	Pred Agent
True Human	500	114
True Agent	201	332

Feature group	n_feat	AUC (alone)	AUC (without)	Delta
interaction	5	**0.751**	0.748	+0.047
gas	6	0.713	0.767	+0.028
approval_security	5	0.656	0.789	+0.005
temporal	7	0.623	0.799	-0.005
all 23	23	**0.794**	—	—

Interaction features (unique_contracts_ratio, top_contract_concentration) are the most important group on the honest set, while temporal features contribute least — removing them actually helps by +0.005. This reverses the leaky-set finding where temporal features dominated via `active_hour_entropy`. The reversal is further evidence that the C1-C4 leakage was concentrated in temporal features.

C1-C4 diagnostic agreement. We applied the C1-C4 heuristic to all 1,147 provenance-labeled addresses. The agreement rate is **48.8%**, below chance. This is a further drop from 55.8% on the v3 set (n=563) and 92.9% on the original 14-address v2 pilot, demonstrating that the C1-C4 heuristic's reliability degrades as the dataset grows and diversifies. As the dataset expands to include Compound V3 liquidators, Chainlink keepers, Gitcoin donors, and PoolTogether depositors, the heuristic's behavioral thresholds increasingly mislabel addresses.

4.1.2 Sensitivity Check: Strict Provenance Subset (n=64, 33 agents, 31 humans). Table 2 reports the strict provenance subset results under LOO-CV and repeated 5x10-CV.

Table 2. Tabular classifier performance on the strict provenance subset (n=64, 33 agents, 31 humans).

Model	Split	AUC-ROC	Precision	Recall	F1	Accu
RandomForest	LOO	**0.7713**	0.7419	0.6970	0.7188	0.718
RandomForest	5x10-CV	**0.8030** +/- 0.102	0.8008	0.7138	0.7363	0.744
GradientBoosting	LOO	0.6276	0.5882	0.6061	0.5970	0.578
GradientBoosting	5x10-CV	0.6940 +/- 0.119	0.6731	0.6638	0.6548	0.649
LogisticRegression	LOO	0.6100	0.6667	0.6061	0.6349	0.640
LogisticRegression	5x10-CV	0.6203 +/- 0.125	0.6974	0.6238	0.6459	0.661
LightGBM	LOO	0.7097	–	–	0.7273	0.718
LightGBM	5x10-CV	0.7040 +/- 0.088	–	–	0.6793	0.676

The n=64 results are broadly consistent with n=1,147: RF is the best tabular model, GBM underperforms, and standard deviations are wide. The slightly lower RF AUC on n=64 (0.7713 LOO vs 0.803 10-fold on n=1,147) may reflect the reduced statistical power of the smaller set rather than a genuine performance difference.

Key findings across both sets.

1. **Random Forest is the best honest tabular model** across both set sizes (10-fold AUC 0.803 on n=1,147, LOO AUC 0.7713 on n=64). 2. **GBM underperforms RF** on both honest sets, despite being nearly identical on the leaky set. We confirm this gap with statistical tests in Section 4.2. 3. **Standard deviations shrink substantially from n=64 to n=1,147** (RF repeated-CV std drops from 0.102 to 0.027), reflecting the benefit of a larger honest corpus. 4. **Feature**

importance shifts. On n=1,147, gas features (gas_price_trailing_zeros_mean, gas_price_round_number_ratio) dominate; on the leaky set, temporal features (active_hour_entropy) dominated. This shift is further evidence of leakage.

Leaky reference (do not cite as headline). On the full 3,316 addresses with C1-C4 labels, GBM achieves 5-fold 5-rep CV AUC 0.9803 +/- 0.006, RF 0.9766 +/- 0.006, LR 0.9521 +/- 0.012, LightGBM 0.9822 +/- 0.006. The gap between 0.98 and 0.80 is the leakage magnitude.

Per-feature univariate AUC (trusted n=64). Table 3 lists the top features by Mann-Whitney-derived AUC.

Table 3. Top univariate AUC on the strict trusted set (n=64).

Rank	Feature	Univariate AUC	Group
1	tx_interval_mean	0.7683	Temporal
2	tx_interval_std	0.7527	Temporal
3	unlimited_approve_ratio	0.7116	Approval
4	unverified_contract_approve_ratio	0.7107	Approval
5	gas_price_round_number_ratio	0.7072	Gas
6	gas_price_trailing_zeros_mean	0.7067	Gas
7	active_hour_entropy	0.7028	Temporal
8	multi_protocol_interaction_count	0.6960	Approval
9	unique_contracts_ratio	0.6950	Interaction
10	method_id_diversity	0.6911	Interaction

Contrast with the leaky set, where `active_hour_entropy` led with AUC 0.9347 and `tx_interval_skewness` reached 0.872. The collapse of these features from 0.93 to 0.70 is direct evidence that the leaky AUCs were inflated by labeling overlap.

4.2 Statistical Significance Tests

Given n=64, we report three complementary tests for each pairwise model comparison: McNemar's test on paired predictions, DeLong's test for paired AUCs, and a 10,000-sample bootstrap CI for the AUC difference.

Table 4. Pairwise statistical tests on the strict trusted set (n=64, LOO predictions).

Comparison	McNemar p	DeLong p	Bootstrap CI95 AUC diff	Significant?
GBM vs RF	**0.0225**	**0.0014**	[-0.24, -0.06]	**Yes** (RF wins)
GBM vs LR	0.4545	0.7895	[-0.11, 0.15]	No
GBM vs Single-Feature	0.7353	<0.001	[-0.52, -0.24]	**Yes**
GBM vs Majority	0.5839	–	–	**No**

Key findings.

1. **Random Forest significantly beats Gradient Boosting** on the trusted set by three independent tests. This is a qualitative reversal of the leaky-set conclusion where GBM and RF were statistically indistinguishable. 2. **GBM is NOT significantly better than a majority classifier** (McNemar p=0.58). Read literally: a model that always predicts AGENT performs as well as the GBM on this set. 3. **A single feature (`tx_interval_mean`) achieves AUC 1.00 at the threshold level** but low accuracy (0.53). This is an artifact of degenerate thresholding on small n; it is reported for completeness but not as a headline claim. The more honest comparison is that RF beats every other model we tried. 4. **Brier scores:** RF 0.194, LR 0.174, LightGBM not reported, GBM 0.355. RF is the best-calibrated model as well as the most accurate.

4.3 Cross-Platform Generalization

Table 5 reports the four cross-platform splits. The critical contrast is between splits 1-2 (cross-label-scheme: provenance vs C1-C4) and splits 3-4 (intra-label-scheme: both C1-C4).

Table 5. Cross-platform transfer AUC.

Split	Direction	GBM AUC	RF AUC	LR AU
Autonolas -> Trusted (honest)	C1-C4 -> provenance	0.269	0.243	0.578
Trusted -> Autonolas	provenance -> C1-C4	0.347	0.341	0.603
Fetch.ai -> AI Arena	C1-C4 -> C1-C4	**0.980**	0.969	0.962
AI Arena -> Fetch.ai	C1-C4 -> C1-C4	**0.992**	0.975	0.973

Key findings.

1. **Cross-label-scheme transfer collapses to worse than chance** (GBM 0.269, RF 0.243). A classifier trained on Autonolas C1-C4 labels and tested on trusted provenance labels produces **anti-predictions**: it labels agents as humans and vice versa. 2. **Intra-label-scheme transfer is near-perfect** (AUC 0.97-0.99). Fetch.ai and AI Arena holders are indistinguishable under the shared C1-C4 oracle, because the oracle is learning the oracle, not the concept. 3. **The 0.27 to 0.99 gap is another expression of leakage.** It tells us the apparent high transfer performance is spurious: it measures consistency of the labeling heuristic, not consistency of behavior across platforms. 4. **Logistic regression, paradoxically, is the best cross-label-scheme model** (0.578 / 0.603). We interpret this as follows: LR cannot fit the leaky labels as tightly as GBM/RF, so it suffers less when the labels change.

Implication. Any future paper claiming high agent-identification AUC on platform-holder data must include this cross-label-scheme transfer test, or the result cannot be distinguished from leakage.

4.4 GNN vs Tabular Classifiers

We train GraphSAGE and GAT on the 3,316-node transaction graph and evaluate on the three splits. Table 6 reports the results.

Table 6. GNN vs tabular AUC across three evaluation tiers.

Model	full_3316 AUC (5-fold)	provenance_1147 AUC (5-fold)	trusted_64 A
GraphSAGE	0.9666 +/- 0.009	–	0.7841 +/- 0.
GAT	0.9341 +/- 0.011	**0.8318 +/- 0.021**	**0.8825 +/- 0**
RF (tabular)	0.9766 +/- 0.006	0.8028 (10-fold)	0.8030 +/- 0.
GBM (tabular)	0.9803 +/- 0.006	0.8095 (10-fold)	0.6940 +/- 0.

Key findings.

1. **GAT is the best honest model at both scales.** On the full provenance set (n=1,147), GAT reaches 5-fold AUC 0.8318 +/- 0.021, beating RF 0.8028 and GBM 0.8095. On the strict subset (n=64), GAT reaches 0.8825 +/- 0.108 vs RF 0.8030. The higher AUC on n=64 is partly due to the smaller, more homogeneous set and partly due to high fold-to-fold variance (std 0.108 vs 0.021 on n=1,147). 2. **GAT variance is dramatically lower on n=1,147.** The std drops from 0.108 (n=64, 13 test nodes per fold) to 0.021 (n=1,147, 229 test nodes per fold), making the n=1,147 result far more reliable.

3. **GraphSAGE does not help on the trusted set.** GraphSAGE 0.7841 (n=64) is close to RF 0.8030 and worse than GAT. We attribute GAT's advantage to the attention mechanism, which can down-weight noisy edges in the sparse transfer graph. 4. **On the leaky full set, the ranking inverts.** GraphSAGE reaches 0.9666 and outperforms GAT 0.9341, while the tabular GBM reaches 0.9803. The leaky ranking has no predictive value for the honest setting. 5. **GNN beats tabular on all honest sets**, which is the first positive empirical result in this paper that survives the leakage correction. We consider this the honest headline: **GAT 5-fold AUC 0.832 on n=1,147** is the primary reference number.

4.5 Combined Paper 1 + Paper 3 Features

Table 7 reports four classifiers on three feature sets: Paper 1 only (23 behavioral features), Paper 3 only (8 AI features), and combined (31 features). The Paper 3 features are described in the companion paper.

Table 7. Classifier AUC by feature set on the strict trusted subset (n=64, LOO and 5x3-CV).

Feature set	Model	LOO AUC	5x3-CV mean AUC
P1 only (23)	GBM	0.6276	0.7267
P1 only (23)	RF	0.7713	0.8183
P1 only (23)	LR	0.6100	0.6696
P1 only (23)	LightGBM	0.7097	0.7040
P3 only (8)	GBM	0.7146	0.7008
P3 only (8)	RF	0.7292	0.7741
P3 only (8)	LR	0.6432	0.6807
P3 only (8)	LightGBM	0.7419	0.7392
Combined (31)	GBM	0.6393	0.7093
Combined (31)	RF	0.7625	0.8151
Combined (31)	LR	0.6256	0.6646
Combined (31)	LightGBM	0.6872	**0.7172**

Key findings.

1. **Combining the two feature sets buys almost nothing on the trusted set.** LightGBM 5x3-CV improves marginally from 0.7040 (P1-only) to 0.7172 (combined), a delta of +0.013. RF actually drops from 0.8183 to 0.8151. 2. **Paper 3 features alone are competitive.** With only 8 features, LightGBM reaches 0.7419 LOO and 0.7392 5x3-CV, close to Paper 1's 23-feature numbers. This is consistent with Paper 3's observation that `hour_entropy` and `behavioral_consistency` are individually strong. 3. **On the full leaky set, combined features look stronger** (GBM 0.9834 vs 0.9787 for P1 only), but this gap is within noise of the leakage artifact and should not be cited. 4. **Implication.** There is no free lunch from naively stacking feature sets on a small honest corpus. Ensemble gains require either more data or smarter late fusion.

4.6 SHAP Feature Explanations

We compute SHAP TreeExplainer values on the RF model trained on n=1,147 to understand feature directionality — not just importance rank but whether higher values push toward agent or human.

Table 8a. Top 10 SHAP mean |value| (RF, n=1,147).

Feature	Mean	SHAP	Direction
top_contract_concentration	0.057	higher → agent	
gas_price_cv	0.044	higher → agent	
unique_contracts_ratio	0.032	higher → agent	
gas_price_round_number_ratio	0.024	higher → agent	
gas_price_trailing_zeros_mean	0.023	higher → agent	
method_id_diversity	0.017	higher → agent	
contract_to_eoa_ratio	0.014	higher → agent	
tx_interval_mean	0.013	higher → agent	
unlimited_approve_ratio	0.013	higher → agent	
sequential_pattern_score	0.013	higher → agent	

Key finding. `active_hour_entropy` — the feature that dominated the leaky set (univariate AUC 0.93) — ranks near the bottom of SHAP importance (0.002) on the honest set. This is the clearest single-feature confirmation that the C1-C4 leakage inflated temporal features. The honest model relies on gas pricing and interaction diversity, not timing patterns.

4.7 Temporal Holdout Evaluation

Random cross-validation allows post-2021 addresses into every training fold, leaking information about the evolving agent population into the model. To measure the true temporal generalization gap, we split the 1,147 provenance-verified addresses by median first-seen block number (block 12,321,026, approximately April 27, 2021). The **train set** consists of 573 addresses first seen before the median block (170 agents, 403 humans; 70% human), and the **test set** consists of 574 addresses first seen after the median block (363 agents, 211 humans; 63% agent). The class-distribution shift—from majority-human in the pre-2021 era to majority-agent in the post-2021 era—reflects the real-world proliferation of on-chain agents.

Table 9. Random CV vs. temporal holdout performance on n=1,147.

Model	Evaluation	AUC	Precision	Recall	F1
RandomForest	Random 5x10-CV	0.803 ± 0.027	0.755	0.623	0.681
RandomForest	Temporal holdout	0.669	0.812	0.309	0.447
	Δ	**−0.134**	+0.057	**−0.314**	−0.234
GradientBoosting	Random 5x10-CV	0.802 ± 0.027	0.725	0.675	0.698
GradientBoosting	Temporal holdout	0.664	0.767	0.499	0.604
	Δ	**−0.137**	+0.042	**−0.176**	−0.094
LogisticRegression	Random 5x10-CV	0.779 ± 0.028	0.711	0.658	0.682
LogisticRegression	Temporal holdout	0.616	0.739	0.422	0.537
	Δ	**−0.163**	+0.028	**−0.236**	−0.145

Key findings.

(1) **Random CV overestimates AUC by 13–16 points.** RF drops from 0.803 to 0.669 ($\Delta = -0.134$), GBM from 0.802 to 0.664 ($\Delta = -0.137$), and LR from 0.779 to 0.616 ($\Delta = -0.163$). This is a substantial and consistent inflation.

(2) **Precision stays high while recall collapses.** RF temporal precision is 0.812 (even higher than random CV), but recall drops from 0.623 to 0.309—the model catches fewer than one in three post-2021 agents. This asymmetry means the classifier is conservative: when it flags an address as an agent, it is usually right, but it misses the majority of new-generation agents.

(3) **GBM is the most temporally robust model** by F1: temporal F1 of 0.604 vs. RF's 0.447 and LR's 0.537. GBM maintains better recall (0.499) than RF (0.309), sacrificing some precision.
(4) **Distribution shift drives the gap.** The training set is 70% human (pre-2021 era, before agent platforms scaled), while the test set is 63% agent (post-2021 era). The model learned a prior that "most addresses are human," which fails in the agent-dominated test period. Additionally, gas-pricing features shift after EIP-1559 (August 2021), and newer agent platforms (Virtuals, AI Arena) use different behavioral patterns than older ones (Autonolas, Fetch.ai).
(5) **This finding strengthens the leakage narrative.** Random CV on provenance-verified labels (0.803) was already 18 points below the leaky AUC (0.98). Now we show that even the honest random-CV number is 13 points above what would be observed in a realistic deployment scenario where the model must generalize forward in time.

4.8 4-Dimensional Security Audit

We run the 4-dimensional audit at three scales: the full 3,302-address leaky set, the n=549 provenance v3 subset (from the 563 set with 14 dropped for empty histories), and the trusted 50-address strict subset. Table 8b reports the n=549 provenance audit alongside the extremes.

Table 8b. Security audit on n=549 provenance set (intermediate tier).

Metric	Agent mean	Human mean	Ratio	p-value
unlimited_approval_rate	0.534	0.439	**1.21x**	**0.0003**
dex_interaction_rate	0.292	0.149	**1.96x**	**<0.0001**
n_unlimited_approvals	97.2	99.8	0.97x	0.232 (NS)
revert_rate	0.041	0.042	0.97x	0.701 (NS)

On the intermediate n=549 set, agents have a significantly higher approval *rate* (53.4% vs 43.9%) and DEX interaction rate (29.2% vs 14.9%), but similar raw approval *counts* (97 vs 100). The extreme 10x ratio from the leaky set is gone; the honest finding is nuanced: agents approve more frequently but don't accumulate more approvals in absolute terms because humans hold approvals longer.

Table 8 below shows the leaky-vs-strict extremes.

Table 8. 4-dimensional security audit: leaky vs honest.

Metric (dimension)	Full 3,302 (C1-C4) agent / human	Agent/human ratio (leaky)	Trusted 50 agent / human	Agent/human ratio (honest)	Direction rev
# approvals (D1)	108.75 / 14.41	**7.55x**	110.09 / 125.11	0.88x	**Yes**
# unlimited approvals (D1)	61.38 / 5.93	**10.35x**	49.14 / 66.14	0.74x	**Yes**
# active approvals (D1)	102.02 / 13.49	7.56x	104.68 / 121.14	0.86x	**Yes**
avg approval age (days, D1)	1170.5 / 1134.9	1.03x	795.6 / 1235.8	0.64x	**Yes**
max approval age (days, D1)	1539.0 / 1295.5	1.19x	1062.2 / 1417.0	0.75x	**Yes**
unlimited approval rate (D1)	0.502 / 0.304	1.65x	0.245 / 0.453	0.54x	**Yes**
DEX interaction rate (D3)	0.228 / 0.133	1.71x	0.071 / 0.041	1.73x	No
swap rate (D3)	0.196 / 0.118	1.66x	0.060 / 0.025	2.39x	No
revert rate (D4)	0.049 / 0.042	1.16x	0.015 / 0.024	0.64x	**Yes**
gas used efficiency (D4)	0.712 / 0.695	1.02x	0.644 / 0.751	0.86x	**Yes**

Key findings.

1. **The agents-have-10x-more-approvals headline does not survive relabeling.** On the trusted set, humans hold **more** unlimited approvals per address than agents (mean 66.1 vs 49.1). This reversal is borderline significant (Mann-Whitney p=0.071 for unlimited approvals, p=0.026 for unlimited approval rate). 2. **Interpretation.** Famous ENS humans (vitalik.eth, Hayden Adams, nick.eth) are heavy long-term DeFi users with many historical approvals. The C1-C4 mining pipeline preferentially labeled token-holder addresses with heavy DeFi footprints as AGENT, which is exactly why the leaky ratio was 10x: we selected for agent-like humans and excluded human-like agents. 3. **DEX interaction rate stays directional** (1.7x on both sets) but the absolute magnitudes drop sharply (0.228 -> 0.071), indicating that the curated-MEV-bot subset is less DEX-active than the platform token holders. 4. **Revert rate reverses:** curated MEV bots have lower revert rates (1.5%) than ENS humans (2.4%), because MEV searchers simulate transactions before broadcasting. 5. **Gas used efficiency reverses:** curated humans are more efficient (0.75) than curated agents (0.64). This is the opposite of the intuitive claim that algorithms are more precise.

The audit direction is not a minor wrinkle. It is a full 180-degree reversal on the headline "agents are riskier than humans on approvals".

4.9 Level 5: Leave-One-Platform-Out (The Cascade Floor)

Levels 1–4 evaluate within-distribution generalization: the training and test sets are drawn from the same pool of platform categories. But a deployment-ready agent detector must recognize agents from platforms it has never seen. We therefore run **leave-one-platform-out cross-validation (LOPO)**: for each of the 14 platform categories with ≥10 addresses, we hold out *all* addresses from that category as the test set and train on the remaining 13 categories. Because each category is single-class (all-agent or all-human), per-platform AUC is undefined; we report per-platform accuracy (which equals recall for agent categories and specificity for human categories) and compute a **pooled AUC** over all 1,131 held-out predictions.

Table 10. Leave-one-platform-out results (GBM, n=1,131 pooled predictions).

Key findings.

(1) **The pooled AUC is 0.42—below random chance.** A coin flip (AUC 0.50) would outperform the classifier. This is not a marginal failure; it is a categorical one. The classifier has learned to distinguish Chainlink keepers from Gelato executors, not agents from humans.
(2) **The best-recognized agent category is pilot agents (80% recall)**, which contains famous, well-studied MEV bots (jared-fromsubway, Wintermute). The classifier recognizes these because their behavioral patterns appear in curated MEV bots and other categories in the training set—a form of overlap, not generalization.
(3) **The worst recognized agent category is defi_hf_trader (14% recall).** These 199 high-frequency Uniswap traders are the largest agent category and the most behaviorally distinctive. When held out, the classifier trained on Chainlink keepers, Gelato executors, and MEV bots does not recognize

Held-out category	n	Class	Accuracy
Agent categories (recall)			
pilot_agent	15	agent	**80.0%**
chainlink_keeper	108	agent	67.6%
gelato_executor	47	agent	61.7%
curated_mev_bot	10	agent	60.0%
keep3r_executor	46	agent	26.1%
expanded_mev_bot	10	agent	20.0%
compound_v3_liquidator	97	agent	15.5%
defi_hf_trader	199	agent	**14.1%**
Human categories (specificity)			
human_exchange_depositor	28	human	**85.7%**
ens_reverse_setter	144	human	76.4%
pooltogether_depositor	89	human	53.9%
expanded_human	25	human	56.0%
gitcoin_donor	146	human	54.8%
human_ens_interaction	167	human	43.1%
Pooled (all 1,131 predictions)			
Pooled AUC	1,131	–	**0.42**
Pooled accuracy	1,131	–	46.4%
Pooled F1	1,131	–	36.9%

them. This proves that "high-frequency DeFi trading" is a platform-specific behavioral mode, not a universal agent signature.

(4) **Compound V3 liquidators are nearly invisible (15.5% recall).** Liquidation bots have a narrow behavioral profile (wait, then execute a single high-gas transaction) that does not resemble the keeper or MEV patterns in the training set.

(5) **Human categories show moderate specificity** (43–86%), with exchange depositors best recognized (86%) and ENS-interaction humans worst (43%). The classifier misclassifies many ENS-active humans as agents, mirroring the heuristic bias that motivated this study.

(6) **Interpretation: the classifier learns platform fingerprints.** Each agent category has a distinctive gas, timing, and interaction profile that separates it from human categories *within* the training distribution. But these profiles are platform-specific, not agent-generic. When the platform is held out, the learned features become noise.

The cascade is now complete. Level 1 (0.98) measured label recovery. Level 2 (0.80) measured within-platform pattern matching. Level 3 (0.67) measured same-era generalization. Level 4 (0.77) confirmed wide uncertainty. Level 5 (0.42) measured true cross-platform generalization—and found none. On-chain agent detection with transaction-level features, as currently formulated, is an open problem.

Important caveat. Every held-out platform category is single-class (all-agent or all-human), which means the LOPO evaluation conflates platform behavioral novelty with label generalization. The 0.42 AUC reflects both genuine inability to learn transferable agent signatures AND the domain shift of encountering an entirely new behavioral profile. A mixed-class LOPO (where some platforms contribute both agents and humans) would isolate generalization more cleanly, but the dataset's provenance structure does not support this design. We therefore interpret Level 5 as measuring *cross-context deployment readiness* rather than pure concept generalization.

4.9.1 Comparison with Published Baselines. Table 11 situates our cascade within published bot-detection results.

Study	Platform	Reported AUC	Label type	Cascade lev
Cresci et al. (2017)	Twitter	0.75–0.85	Crowd-sourced	∼Level 2
Varol et al. (2017)	Reddit	∼0.80	API flags	∼Level 2
Chen et al. (2020)	Ethereum	0.92	Heuristic	∼Level 1
This work	Ethereum	0.98 / 0.80 / 0.67 / 0.42	Cascade L1–L5	All five

No prior blockchain bot-detection paper reports a Level 5 evaluation. Until they do, their high-AUC numbers cannot be distinguished from platform fingerprinting.

5 Threats to Validity: The C1-C4 Label Leakage Discovery

This section is the methodological centerpiece of the paper and documents the leakage discovery, the forensic evidence, the fix, and the implications.

5.1 The Bug

In our labeling script `paper1_onchain_agent_id/features/verify_c1c4.py`, the `_check_c3` function gates AGENT labels on three quantities:

- `tx_interval_cv` (coefficient of variation of inter-transaction intervals) - `hour_entropy` (Shannon entropy over UTC hour bucketing) - `burst_ratio` (fraction of consecutive transactions within 10 s)

These three quantities are algebraically the same as three of the top classifier features in the 23-feature vector:

- `tx_interval_std = tx_interval_mean x tx_interval_cv` (entangled with feature 2) - `active_hour_entropy = hour_entropy` (identical to feature 4) - `burst_frequency = burst_ratio` (identical to feature 7)

Any address that was labeled AGENT because `hour_entropy > 2.5` is then used to train a classifier whose top feature is `active_hour_entropy`. The model is **guaranteed** to re-derive the labeling rule from the features.

5.2 The Evidence

Evidence 1: The honest vs leaky AUC gap.

Metric	Leaky (C1-C4, n=3316)	Full provenance (n=1,147)
GBM AUC	0.9803	0.8095 (10-fold)
RF AUC	0.9766	**0.8028** (10-fold)
GAT AUC	0.9341	**0.8318** (5-fold)
Top feature univariate AUC	active_hour_entropy: 0.9347	gas_price_trailing_zeros_n

The AUC drops from 0.98 (leaky) to 0.80 (full provenance) to 0.77 (strict provenance) as we progressively tighten the labeling purity. The top-feature ranking also shifts: on the leaky set, the leaked temporal feature `active_hour_entropy` dominated; on n=1,147, gas features dominate; and on n=64, the temporal feature `tx_interval_mean`

leads but with a much lower AUC (0.77 vs 0.93). This is the signature of label leakage.

Evidence 2: Inter-label agreement is low and worsens at scale.

We apply the GBM trained on the 64 provenance labels transductively to the 3,252 platform-token-holder rows (whose C1-C4 labels we no longer trust). The inter-label agreement between the honest prediction and the C1-C4 label is **37.4%**, which is below random chance for balanced binary classification. The honest classifier does not endorse the majority of C1-C4 labels.

Furthermore, applying the C1-C4 heuristic directly to the 1,147 full provenance addresses yields a diagnostic agreement of only **48.8%**, below chance. This was 55.8% on the v3 set (n=563) and 92.9% on the original 14-address v2 pilot, demonstrating that the C1-C4 heuristic's reliability degrades rapidly as the dataset diversifies beyond the narrow pilot categories. This is a sixth piece of evidence against heuristic labeling.

Evidence 3: Statistical tests on n=64 reveal structural cracks.

- RF significantly beats GBM (McNemar p=0.0225, DeLong p=0.0014, bootstrap CI95 [-0.24, -0.06]). - GBM vs Majority: NOT significantly different (GBM LOO accuracy 0.578 vs Majority 0.516, McNemar p=0.58). - Single best feature matches GBM.

These are the three symptoms you expect to see when a classifier's performance is being propped up by a leaking label: the strongest model (GBM) behaves like a majority baseline once the leak is sealed.

Evidence 4: Cross-platform transfer reverses sign.

Train on Autonolas C1-C4 labels and test on trusted provenance: AUC 0.24-0.34. Train on Fetch.ai C1-C4 labels and test on AI Arena C1-C4 labels: AUC 0.97-0.99. The gap is a direct measurement of the labeling-scheme-specific signal that the classifier picked up.

Evidence 5: Security audit direction reverses.

The "agents hold 10x more unlimited approvals" headline from the leaky set becomes "humans hold 1.3x more unlimited approvals" on the trusted set. The direction of a 4-dimensional audit is not supposed to depend on labeling heuristic. It does here because the heuristic preferentially admits heavy-DeFi users as AGENT.

5.3 The Fix

We decided on the following:

1. **Demote C1-C4 from a labeling oracle to a downstream diagnostic tool.** It can still be used to rank how likely an address is to be an agent given a trained classifier, but it cannot be the source of ground truth. 2. **Expand the provenance-labeled set from 64 to 1,147.** Through systematic v3 and v4 mining campaigns, we grew the honest corpus from 64 to 1,147 addresses across 18 provenance source categories, reducing the statistical uncertainty and enabling 10-fold CV instead of LOO. 3. **Retain n=64 as a strict sensitivity check.** The original 64 provenance-only addresses serve as a conservative reference point. 4. **Apply the RF trained on the 1,147 provenance labels transductively** to the 3,252 platform-token-holder rows to generate honest predictions, and publish these predictions as weak labels for future re-mining rather than as ground truth. 5. **Report leaky, full-provenance, and strict-provenance numbers** throughout the paper with clear labels. 6. **Publish the leakage case study** as Section 5 of this paper.

Algorithm 1: Three-Step Leakage Audit for Behavioral Classifiers

Input: Feature set $\mathcal{F}$, labeling pipeline $\mathcal{L}$, classifier h
Output: Leakage verdict: CONFIRMED / NOT DETECTED
Step 1: Label–feature overlap check.;
Let $\mathcal{L}_q$ = all quantities used in $\mathcal{L}$ (thresholds, filters, derived scores);
foreach $f \in \mathcal{F}$ **do**
  Compute Jaccard similarity $J(f, \ell)$ for each $\ell \in \mathcal{L}_q$, allowing algebraic transforms (log, ratio, CV ↔ std/mean, renaming);
  if $J(f, \ell) > 0$ *or* $|Pearson(f, \ell)| > 0.9$ **then**
    Flag (f, ℓ) as entangled;
  end
end
if *no flags* **then**
  return NOT DETECTED (but proceed to Steps 2–3 as confirmation);
end
Step 2: AUC across label-purity tiers.;
Define tiers: T_1 = heuristic labels (leaky), T_2 = provenance labels, T_3 = strict provenance;
Train h on each tier; record AUC: a_1, a_2, a_3;
if $a_1 > a_2 > a_3$ *(monotonic decline with purity)* **then**
  Label-quality sensitivity confirmed;
end
Step 3: Cross-label-scheme transfer.;
Train h on T_1 labels, test on T_2 labels (and vice versa);
if *cross-scheme AUC* < 0.5 *(worse than chance)* **then**
  The classifier learned the labeling scheme, not the concept;
  return CONFIRMED;
end
return NOT DETECTED;

5.4 A Reusable Leakage-Audit Toolkit

The leakage we found is subtle: none of the classifier features are literally one of the gating quantities. The leak is through algebraic identities (`tx_interval_std` ≈ `tx_interval_cv` × `tx_interval_mean`) and name collisions (`hour_entropy` gated the label, `active_hour_entropy` is a feature). A reviewer running a superficial check would not catch it; neither did we for several weeks.

We distill our experience into a three-step audit that any behavioral-classifier study on blockchain data can apply. Algorithm 1 formalizes the procedure; if AUC drops monotonically across label-purity tiers **and** cross-scheme transfer collapses, leakage is confirmed.

Application to this study. Step 1 identified three entangled pairs (`hour_entropy`/`active_hour_entropy`, `burst_ratio`/`burst_frequency`, `tx_interval_cv`/`tx_interval_std`). Step 2 confirmed monotonic AUC decline: 0.98 (leaky) → 0.80 (full provenance) → 0.77 (strict provenance). Step 3 showed Autonolas→trusted transfer AUC of 0.24–0.34, worse than chance. All three steps converge: leakage confirmed.

We release the implementing scripts (`verify_c1c4.py`, `pipeline_provenance.py`, `statistical_tests.py`, `cross_platform_eval.py`) as a reusable toolkit that other researchers can adapt to their own labeling pipelines.

5.5 Other Threats

Beyond the C1-C4 issue, several additional threats apply.

- **`defi_hf_trader` label quality.** The largest agent category in the full provenance set is `defi_hf_trader` (199 addresses, 37% of all agents), labeled as "agent" because the address makes >50 Uniswap router calls per active day. While this threshold strongly suggests automated execution, it is a behavioral criterion rather than a provenance-pure label. Some of these addresses could be humans using aggressive trading interfaces (e.g., custom front-ends with auto-routing). If a significant fraction are misclassified, the n=1,147 AUC would be artificially depressed (because the classifier is being penalized for correctly identifying humans that we mislabeled as agents) or inflated (if the behavioral criterion leaks into features the way C1-C4 did). We note that the >50-daily-calls threshold is at least one step removed from the 23 classifier features: none of the features directly encode daily Uniswap call count. However, `top_contract_concentration` and `method_id_diversity` are correlated with high-frequency Uniswap usage, so a residual leak is possible. This is the most important caveat on the n=1,147 results and should be investigated in future work by validating a random sample of `defi_hf_trader` addresses against deployment metadata or operator disclosures. - **Sample size on the strict set.** n=64 is too small to support fine-grained claims. 5-fold CV has 13 test samples per fold; AUC confidence intervals are wide. The full provenance set (n=1,147) substantially mitigates this concern for the primary results. - **Selection bias.** The strict set's 33 agents and 31 humans are public, named, famous addresses. The expanded set diversifies the sources but still over-represents high-frequency DeFi traders among agents and ENS-active users among humans. - **Feature drift.** Agent behavior evolves as platforms update. Our features reflect a specific time window. - **Adversarial evasion.** An adversary aware of the feature set can mimic human behavior by injecting jitter into gas pricing and transaction timing. - **Chain coverage.** We only study Ethereum mainnet. Agents on Base, Arbitrum, Solana, and other chains are out of scope. - **Contract-type confounding.** We excluded router contracts and exchange hot wallets from training, but edge cases remain.

6 Discussion: What Does the Cascade Mean?

6.1 For Researchers

The cascade from 0.98 to 0.42 has a clear methodological implication: **every published blockchain bot-detection result needs a Level 5 evaluation.** Without holding out entire platform categories, a reported AUC cannot be distinguished from platform fingerprinting. The field's standard practice—random cross-validation on pooled multi-source data—corresponds to Level 2 at best, which we have shown overstates deployment-realistic performance by at least 38 AUC points relative to the cross-platform test.

We recommend that future work report all five cascade levels (or justify why a level is inapplicable) and treat Level 5 as the primary generalization metric. The three-step leakage-audit toolkit (Algorithm 1) provides a lightweight sanity check that can be applied before any cross-validation.

6.2 For Practitioners

The cascade reveals a sharp boundary between what works and what does not:

- **Within-platform triage works (Level 2).** At AUC 0.80 and precision 0.81, the classifier is useful for flagging high-confidence agent candidates within a known platform distribution. Security auditors can deploy it as a triage layer with a false-positive rate under 20%.
- **Cross-platform deployment does not work (Level 5).** At pooled AUC 0.42, the classifier produces anti-predictions on unseen platform categories. It should not be used to estimate the fraction of automated activity across the Ethereum ecosystem, because the false-negative rate on novel agent categories exceeds 85%.
- **Temporal drift is real (Level 3).** Even within known platforms, the 13-point AUC gap between random CV and temporal holdout means the classifier degrades within months. Periodic retraining with provenance-verified addresses from newly launched platforms is essential.

6.3 For Regulators

The Level 5 result has a direct policy implication: **behavioral monitoring alone cannot reliably distinguish AI agents from humans across operational contexts.** Any regulatory framework that depends on transaction-level behavioral classification to identify automated activity will systematically miss novel agent categories while over-flagging certain human behavioral profiles (e.g., ENS-active DeFi users). Until agent detection generalizes cross-platform, complementary approaches—platform registries, voluntary disclosures, EIP-7702 delegation signals—are necessary.

6.4 Limitations

(1) **Source composition bias.** Agents are dominated by high-frequency DeFi traders (37% of agent addresses); humans are dominated by ENS-interaction-verified accounts (27% of human addresses). The classifier partly learns "high-frequency Uniswap trader vs. casual ENS holder" rather than a general agent-vs-human distinction—and the Level 5 result confirms this is exactly what it learns.
(2) **The `defi_hf_trader` label quality.** The largest agent category (199 addresses) uses a behavioral threshold (>50 daily Uniswap calls) that is not strictly provenance-pure. Some addresses may be humans using aggressive trading interfaces. However, this category's 14% recall under LOPO suggests the classifier does not rely on it for generalization.
(3) **Single-class platforms preclude per-platform AUC.** Because each platform category is either all-agent or all-human, we cannot compute per-platform AUC, only per-platform accuracy. The pooled AUC aggregates across all held-out predictions.
(4) **Chain coverage.** We study Ethereum mainnet only. Agents on Base, Arbitrum, Solana, and other chains are out of scope.

(5) **The security audit is small-sample.** The honest direction reversal is based on 22 agents and 28 humans. We claim the 10x leaky-set ratio was an artifact, not that the reversal necessarily holds at scale.

6.5 Future Directions

- **Platform-invariant features.** The Level 5 failure implies that the 23-feature set captures platform-specific behavioral modes. Future work should explore features that are invariant across platforms: graph-topological properties, cross-protocol interaction patterns, or latent representations learned via contrastive pre-training across platform categories.
- **EIP-7702 signals.** EIP-7702 delegation creates a new observable that could distinguish agent wallets without behavioral classification.
- **Cross-chain evaluation.** Extending the cascade to multi-chain agents (Base, Arbitrum) would test whether cross-chain behavioral features generalize better than single-chain ones.
- **Adversarial stress tests.** Inject behavioral noise into agent transactions to measure robustness against evasion.
- **Rolling temporal splits.** Evaluate temporal degradation at multiple cutoff dates to establish a retraining cadence.

7 Conclusion

The cascade from 0.98 to 0.42 is itself our main contribution. It demonstrates that on-chain AI agent detection, as currently formulated with 23 transaction-level behavioral features, does not generalize beyond the platforms used for training.

At each level of the cascade, we stripped away an evaluation shortcut and lost performance:

- **Level 1 → 2** (0.98 → 0.80): Fixing label leakage removed 18 AUC points.
- **Level 2 → 3** (0.80 → 0.67): A temporal holdout removed 13 more points.
- **Level 3 → 5** (0.67 → 0.42): Holding out entire platforms removed the remaining signal and pushed AUC below chance.

The Level 5 result is the decisive finding. A pooled AUC of 0.42 across 14 held-out platform categories means the classifier has learned to distinguish Chainlink keepers from Gelato executors—not agents from humans. The best-recognized category (pilot agents, 80% recall) succeeds because similar addresses exist in other training categories; the worst (DeFi high-frequency traders, 14% recall) fails because its behavioral profile is unique. This is platform fingerprinting, not agent detection.

Along the way, we discovered that the widely assumed "agents are riskier" narrative is a labeling artifact (humans hold more unlimited approvals under honest labels), and we formalized a reusable three-step leakage-audit toolkit (Algorithm 1) that other blockchain-ML researchers can apply.

For the field to make progress on this problem, three things must change: (1) cross-platform evaluation (Level 5) must become a non-negotiable reporting standard; (2) feature engineering must move toward platform-invariant representations; and (3) the community must accept that honest AUCs will be lower than leaky ones—and that lower, honest numbers are the only ones worth building on. All code, features, labels, and diagnostic artifacts are released.

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